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Appendix

A Multiple Watermark Detection

Setup and Notation

– n: Total number of bits in the hidden message $(m_1, \ldots, m_n) \in \{0,1\}^n$.

- k: Number of 1-bits in the message, i.e., $k = \sum_{i=1}^{n} m_i$.
- T: Total number of tokens generated (i.e., the covertext length).
- δ: Strength of the watermark perturbation. The probability of PRF-labeled "1" tokens is shifted from $p_0 = 0.5$ to
$$p_w = \frac{2(1+\delta)}{4+\delta}, \quad \text{where } p_w > 0.5.$$
- Key usage rule: if $m_i = 1$, then key i is selected with probability $1/k$ (uniformly among the 1-bits). If $m_i = 0$, then key i is never used.

Distribution of Tokens Labeled "1"

For each message bit index i, define s_i as the number of tokens in the final covertext that are labeled "1" by the PRF. Then:

$$X_i \;=\; \frac{s_i}{T} \quad \text{is the fraction of tokens labeled 1 for bit } i.$$

Case 1: If $m_i = 0$, it is never chosen. Hence the fraction of tokens labeled "1" is exactly

$$0.5 \quad \text{(no watermark shift)}.$$

That is,

$$\mathbb{E}[X_i] \;=\; 0.5.$$

Case 2: If $m_i = 1$, it is chosen with probability $1/k$. Whenever chosen, the fraction of tokens labeled 1 is p_w, otherwise it remains 0.5. Thus:

$$\mathbb{E}[X_i] \;=\; \frac{1}{k}\,p_w \;+\; \Big(1 - \frac{1}{k}\Big)\,(0.5) \;=\; 0.5 + \frac{1}{k}\,(p_w - 0.5).$$

Binomial Z-Test for Detection

Under the null hypothesis ($m_i = 0$), $s_i \sim \text{Binomial}\big(T, 0.5\big)$ approximately. Hence

$$\mathbb{E}[X_i] = 0.5, \quad \text{Var}(X_i) = \frac{0.25}{T}.$$

Define the Z-score for detecting a deviation:

$$Z_i \;=\; \frac{X_i - 0.5}{\sqrt{0.25/T}} \;=\; 2\sqrt{T}\,\big(X_i - 0.5\big).$$

Under the alternative hypothesis ($m_i = 1$), the expected deviation is

$$\Delta \;=\; \frac{1}{k}\,\big(p_w - 0.5\big).$$

So,

$$\mathbb{E}[X_i] = 0.5 + \Delta \quad \text{and} \quad \mathbb{E}[Z_i] \;=\; 2\sqrt{T}\,\Delta \;=\; 2\sqrt{T}\,\frac{\big(p_w - 0.5\big)}{k}.$$

Let z_{th} be the one-sided detection threshold. We conclude $m_i = 1$ if

$$Z_i \;>\; z_{\text{th}}, \quad \text{and otherwise conclude } m_i = 0.$$

Under the null hypothesis ($m_i = 0$), Z_i is approximately standard normal, so

$$\mathbb{P}[\text{false positive}] \;\approx\; 1 - \Phi(z_{\text{th}}).$$

Under the alternative hypothesis ($m_i = 1$), the Z-score mean shifts upward; with large T, the shift exceeds z_{th} with high probability.

Minimum Covertext Length T for Reliable Detection

To ensure per-bit false positive and false negative rates of at most α, set:

$$z_{\text{th}} \;=\; \Phi^{-1}(1-\alpha).$$

Reliable detection requires

$$\mathbb{E}[Z_i] \;>\; z_{\text{th}},$$

which implies

$$T \;>\; \frac{z_{\text{th}}^2\, k^2}{4\left(p_w - 0.5\right)^2}.$$

This ensures that each *1*-bit surpasses the threshold, while each *0*-bit does not with high probability.

Multi-Bit Recovery

To recover all n bits with overall error probability $\leq \epsilon$, we can set each bit's $\alpha_{\text{bit}} = \epsilon/n$. Then

$$z_{\text{th}} \;=\; \Phi^{-1}\!\left(1 - \tfrac{\epsilon}{n}\right).$$

In the worst case ($k \leq n$), set $k_{\max} = n$. Then a sufficient bound is

$$T \;>\; \frac{z_{\text{th}}^2\left(k_{\max}\right)^2}{4\left(p_w - 0.5\right)^2} \;=\; \frac{z_{\text{th}}^2\, n^2}{4\,(p_w - 0.5)^2}.$$

This ensures that for any actual $k \leq n$, each *1*-bit will produce a Z-score above z_{th}, and *0*-bits will not exceed z_{th} with high probability.

Example

- Let $n = 3$, $\delta = 0.1 \implies p_w = \frac{2(1+\delta)}{4+\delta} = \frac{2.2}{4.1} \approx 0.5366$.
- Then $p_w - 0.5 \approx 0.0366$.
- For 95% success ($\epsilon = 0.05$), $\implies \alpha_{\text{bit}} = 0.05/3 \approx 0.0167$, $\implies z_{\text{th}} \approx 2.17$.
- Worst-case $k_{\max} = 3$:

$$T \;>\; \frac{2.17^2 \times 3^2}{4\,(0.0366)^2} \;\approx\; 7900,$$

  Thus, a covertext length of approximately 8000 tokens suffices to recover all bits with high probability.